

Who Can Still Afford to Live Here?

How Remote Work Reshaped Housing Affordability Across the United States

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1. Overview

1.1 The Problem

The rise of remote work has reshaped migration patterns across the United States, enabling higher-income remote workers to relocate away from traditional urban job centers to more affordable suburban neighborhoods. This inflow of remote workers can lead to upward pressure on local housing markets, increasing costs and reducing affordability for existing local residents. This dynamic has also generated uneven regional impacts across metropolitan areas.

While some large metropolitan cores experienced slower rent growth or population decline, surrounding suburban cities face intensified competition for their housing stock. These shifts, commonly referred to as the post-pandemic “Donut Effect”, have led to patterns of displacement, growing concerns towards long-term housing affordability, and increased strain on local infrastructure. Such shifts can also create spillover effects, in which nearby areas experience rent and home price increases even if they are not directly receiving new residents.

In the San Francisco Bay Area, these trends have intensified in suburban neighborhoods, where an influx of remote workers since the COVID-19 pandemic has contributed to rising rent and home prices and increased affordability pressures for local residents. Our experiment will measure the effects of implementing a dynamic local income tax (DLIT) on cities surrounding San Francisco’s CBD. Specifically, we will examine whether this intervention reduces the incentive for high-income remote workers to relocate from San Francisco to the surrounding cities, and whether moderating this inflow helps stabilize rent and housing costs for existing residents.

The results are intended for use to help local policymakers manage remote-worker migration pressures while protecting housing affordability for existing residents. By

determining whether a DLIT can mitigate inflows of remote-workers into surrounding cities, this research provides an evidence-based tool for designing policies that manage displacement risks and support long-term regional housing stability.

1.2 Intended Audience

The primary intended audience for our study is the Bay Area Housing Finance Authority (BAHFA), the regional agency responsible for funding, preserving, and producing affordable housing across the San Francisco Bay Area's nine counties and 101 cities. BAHFA's mission stands on the 3P's framework – Production, Preservation, and Protection. They work to invest in the preservation of existing affordable housing as well as the construction of new affordable housing in order to protect low- and middle-income residents from displacement from their communities. This intervention directly advances these Protection and Preservation pillars by stabilizing housing prices and reducing displacement pressures for existing residents. Because BAHFA was created specifically to address housing challenges greater than any one city or county could address, it serves as the most reasonable governing body for responding to the regional phenomenon known as the “Donut Effect”. BAHFA was created precisely to coordinate solutions across jurisdictions, and therefore, is the intended audience best positioned to evaluate, implement, and enforce this dynamic taxing intervention.

A secondary audience of interest is the San Francisco Mayor's Office of Housing and Community Development (MOHCD), which stands to benefit from reversing the impact of the “Donut Effect”, which prompted the migration of high-income residents out of the city. Retaining higher-income residents is important to the city of San Francisco not only to keep neighborhood stability but also because it is essential for sustaining the city's tax base which supports public services & safety, administration, and infrastructure.

Our tertiary intended audience is the low- and middle- income residents of the Bay Area, who are the most susceptible to rising housing costs and displacement pressures intensified by the “Donut Effect”. These households struggle to compete with the higher-income remote workers relocating into their suburbs, making them disproportionately vulnerable to being priced out of their own communities.

Our intervention is designed to shield these residents from the negative impacts of the “Donut Effect” by reducing the affordability pressures they face in local housing markets. Since this intervention only applies to new, higher income residents, it avoids burdening current households by stabilizing housing costs and preventing displacement of those unable to keep up with rising costs of living.

1.3 Existing Literature

The widespread adoption of remote work, accelerated by the COVID-19 pandemic, has fundamentally transformed urban structures, housing markets, and demographic trends in the United States (National Bureau of Economic Research, 2022). This shift, called the “Donut Effect”, has driven significant changes in housing demand, contributing to notable house price appreciation, particularly as resources have reallocated away from dense Central Business Districts (CBD) toward lower-density suburban areas, resulting in slower rent and housing price growth in CBDs (Levin, 2025).

While remote work has encouraged migration toward smaller metropolitan areas, most departures from large city centers remained within the broader metropolitan regions, with many households relocating to surrounding suburbs (Ramani, 2021). The city of San Francisco alone experienced a population loss of 6.5% from 2020 to 2023, while the larger Bay Area experienced a population loss of 2.6%. The migration outflow of the SF central business district and surrounding Bay Area is higher than the migration outflow of 1.8% in the State of California during the same period (Johnson, 2023).

Research gaps remain around understanding the long-term geographic and economic effects of remote work, especially regarding the downstream impacts of the “Donut Effect” (Gupta, 2022). Additionally, ongoing demographic shifts, particularly pandemic-induced migration changes and their influence on regional productivity pose critical questions for future housing demand and real estate markets (Tilstra, 2024).

Addressing these complex and evolving trends will require further research to anticipate how remote work will reshape spatial economic dynamics, affordability, and urban development in the post-pandemic era (Weingarden, 2022). The majority of state and local tax is generated from low and middle-income families, and there is currently a lack of bracketed local income tax across the US (Institute on Taxation and Economic Policy, 2024). This study adopts an experimental framework to test the causal impact of dynamically adjusted local income tax interventions on high-income households moving into the surrounding cities neighboring San Francisco.

1.4 Anticipated Impact

This research intends to provide causal evidence on whether implementing a dynamically adjusted local income tax can stabilize housing affordability for surrounding cities by reducing migration outflows from San Francisco, thus reversing the “Donut Effect”. As remote-workers (who are typically higher-income workers) move into surrounding, more affordable cities they unintentionally raise rent and home prices as well as the overall cost of living for existing residents of the area.

By targeting only high-earning newcomers rather than existing households, this policy is designed to slow the rapid influx of remote workers into the surrounding suburban cities and preserve affordability for current residents. This DLIT serves as a corrective mechanism – newcomers earning more than the median household income of the city are taxed accordingly, thus reducing the incentive to relocate solely to capitalize on the comparatively low housing costs in these surrounding cities.

The revenue generated through these taxes can be reinvested into local affordability measures such as rental assistance programs, affordable housing preservation, and supporting infrastructure improvements and creation. Ultimately, the anticipated impact of this intervention is a more balanced regional housing market, where affordability is preserved for existing residents, displacement pressures are reduced, and cities can retain the tax base required to support essential public services.

The results of this study will help policymakers (such as BAHFA) determine whether DLIT can meaningfully counteract the uneven regional pressures created by the rise in remote work. If successful, this intervention could serve as a new policy mechanism for protecting affordability in vulnerable communities facing similar challenges.

2. Research Question

2.1 Main Research Question

Does the introduction of a dynamic local income tax (DLIT) in Bay Area cities surrounding San Francisco reduce migration flows out of the central business district therefore decreasing the “Donut Effect”?

2.2 Sub-Questions

1. Do we see higher migration outflows from SF into surrounding cities that do not have a DLIT compared to the cities that do?
2. Does the dynamic local income tax (DLIT) slow the rise of rent and home prices in surrounding Bay Area treatment cities versus control cities?

2.3 Definitions

Dynamic local income tax (DLIT) - Dynamically adjusted local income tax on incoming residents who move into the area after the start of the research.

Bay Area cities - Daly City, South San Francisco, Brisbane, Pacifica, Colma, Oakland, Berkeley, Alameda, San Bruno, San Rafael, Sausalito, Mill Valley.

Amenities - Desirable features and services that improve residents' quality of life [education & learning, transportation, shopping & services, safety & governance, and healthcare].

Central Business District (CBD) - The main business and commercial area of a town or city.

Donut Effect - Where central business districts became less populated while suburban rings experienced growth.

Higher-income household - Households more than 1 standard deviation from the median-household income of the particular surrounding city.

Middle-income household - Households earning +/- 1 standard deviation from the median-household income of the particular surrounding city.

Low-income household - Households earning less than 1 standard deviation from the median-household income of the particular surrounding city.

Median Household income - Median household income at the time of the start of the study (Jan 1st, 2026) reflective of the previous 2025 tax year.

3. Study Design

3.1 Background for experiment:

High-income households in San Francisco are increasingly migrating to surrounding lower-cost cities, exacerbating housing shortages and accelerating cost pressures in those areas. The outflows from the city center, as defined in the "Donut Effect", raises questions about the unintended consequences of that regional migration.

The purpose of this experiment is to study the effects of imposing a dynamic local income tax on high-income households relocating from San Francisco's city center to the surrounding Bay Area cities, specifically addressing changes in the net migration and housing costs.

Prior studies on migration economics, such as those like Lee's Theory of Migration, indicate that financial incentives can drive high-income earners to seek better social and economic benefits by relocation to less expensive areas, as in the case with the surrounding Bay Area cities. This study tests the dynamic local income tax as an

intervention to curb the migration from the CBD to the surrounding cities by minimizing the financial benefit of high-income households seeking financial arbitrage.

3.2 Review Theory:

We're predicting that the migration from San Francisco to surrounding lower-income areas will decrease, minimize rent and house pricing increases. By dynamically adjusting the local income tax structure, people with higher household incomes will not be incentivized to move away from the CBD of San Francisco to seek financial arbitrage from a lower cost of living area in the surrounding cities. Changing the tax structure; prices will stabilize, there won't be so much demand based on the reduced migration. Essentially stabilizing the donut.

Table: Expectation on How will treatment Affect Migration and Housing Costs

	Control Group (no DLIT)	Treatment Group (DLIT)
Migration	Negligible difference	Reduce inflow into surrounding cities
Housing and Rent Cost	Housing and rent costs to rise faster than the city center	Minimize rent and housing cost increases

3.3 Broad Experiment Context:

The experiment will take place in the twelve cities surrounding San Francisco. The treatment will be applied as a local income tax to new residents of those surrounding cities.

- Who: This experiment targets the 12 cities surrounding San Francisco, CA. Daly City, South San Francisco, Brisbane, Pacifica, Colma, Oakland, Berkeley, Alameda, San Bruno, San Rafael, Sausalito, Mill Valley.
- What: The intervention is applying a DLIT to the surrounding cities.
- When: The experiment will take place starting Jan 1st, 2026 and last for five years until Dec 31st, 2031.

Example of a dynamic local income tax bracket applied to a surrounding city. Each surrounding city will have a different dynamic tax system based on the median household income of that city.

2026 Local Income Tax Rates and Brackets		
Tax Rate	on taxable income from...	up to...
0.00%	\$0.00	\$107,799.00
5.00%	\$108,000.00	\$120,844.83
7.00%	\$120,845.00	\$133,689.83
10.00%	\$133,690.00	\$146,534.83
12.00%	\$146,538.00	And up

4. Data

Our study will generate its data through a policy-based field experiment in which a DLIT is implemented in the selected “treatment” cities. This tax is only applied to new residents with incomes above the median household income of the area. The purpose of the tax is to assess whether reducing the incentive for high-income remote workers to relocate into these cities moderates the rent growth and stabilizes housing affordability for existing residents. The data produced by this intervention will include:

- Number of new household move-ins by income bracket.
- Changes in home and rent price.
- Changes in migration flows from San Francisco into treatment vs control cities.
- Tax revenue generated by DLIT.

These outcome variables will allow us to measure how the DLIT influences migration patterns and housing affordability.

4.1 Preliminary Observational Research

Prior to implementing the experiment, we will conduct observational baseline research using existing data to understand the pre-intervention conditions across all cities.

Pre-intervention data will include:

- Historical rent and home price trends
- Migration flows between San Francisco and surrounding cities
- Household income distributions
- Housing supply and vacancy rates
- Share of remote work occupations

Because this data already exists and is readily available, we do not need to collect new data. The “pain” of collecting new data would not generate meaningful gain, as existing data is substantial enough to inform our experimental design.

The publicly available data:

- U.S. Census Bureau & American Community Survey – economic and migration statistics
- Zillow & Redfin – housing and rental price indices
- Bureau of Labor Statistics – occupation data to estimate remote work prevalence

Together, this data will provide a comprehensive view of the pre-intervention patterns needed to design the experiment and conduct casual analysis.

4.2 Data on Covariates

To isolate the causal effect of the DLIT, we will incorporate a set of city-level covariates known to influence both housing markets and migration patterns. These variables account for structural differences between treatment and control cities.

Existing Covariate Data (matched with experimental outcomes)

- Median rent & home price
- Median household income
- Vacancy rates
- Share of remote-work occupations (obtained through BLS & tenant application)
- Distance & commute time to San Francisco
- Local amenities

These covariates will be matched to the experimental outcome data we will collect at the end of the experiment.

4.3 Data Collection During Experiment

After the five-year intervention period, we will collect new data generated by the policy implementation. This allows us to study how the treatment affects migration flows and downstream housing costs.

Data collected after the experiment includes:

- Count of new move-ins by income bracket (verified through local tax filings)
- Changes in rent and home prices
- Tax revenue generated by each bracket
- Share of remote-work occupations (obtained through BLS & tenant application)

This data will be collected from local government or tax authorities, housing agencies, and public records that track migration, income, and occupation statistics. These are administrative datasets, they provide high accuracy and minimal barriers to response.

4.4 Intervention

- Treatment Condition: A DLIT for out-of-city residents who move into the surrounding city during the duration of the experiment. Only a single treatment will be applied to the cities in the sampling frame, but the tax brackets will be based on the median income of the individual cities themselves.
- Control Condition: No DLIT applied to a city surrounding SF.

5. Sample

Population: The population of interest is the 100 Bay Area cities surrounding San Francisco.

Sampling frame: The sampling frame is the 12 nearest cities surrounding San Francisco from the 100 total cities in the Bay Area. The 12 specific cities were chosen based on their proximity to the city center of San Francisco.

12 cities of interest: Daly City, South San Francisco, Brisbane, Pacifica, Colma, Oakland, Berkeley, Alameda, San Bruno, San Rafael, Sausalito, Mill Valley.

Sampling technique: Each city will be assigned a randomly generated number. Then those cities will be sorted based on ascending order, split in half for control vs. treatment.

- Sample Size: 12 cities surrounding San Francisco
- Size of treatment & control group:

	Control	Treatment
12 Sampled Cities	N = 6	N = 6

Randomization:

The application of the treatment to the 12 surrounding cities follows a simple random assignment. Each city will receive a randomly generated number from a number generator. The surrounding cities will be sorted in ascending order based on their associated randomly generated number. The first six cities will apply the treatment of a dynamic local income tax. The remaining six cities will be the control cities in the study.

6. Hypotheses

Main Research Question

Null Hypothesis (H_0a): The introduction of a dynamic local income tax yields no significant effect on the population outflows from the city center into the surrounding cities.

Alternative Hypothesis (H_1a): The introduction of a dynamic local income tax significantly decreased the population outflows from the city center into the surrounding cities.

Sub-Question

Null Hypothesis (H_0b): The introduction of a dynamic local income tax yields no significant effect on the increase of rent and housing costs.

Alternative Hypothesis (H_1b): The introduction of a dynamic local income tax will significantly decrease the rise of rent and housing costs.

7. Variables

Covariate Data (IV)

1. Median Household Income
 - Help capture the baseline economic profile of each city.
 - Essential for defining “high-income households”.
2. Median Rent Price
 - Captures baseline differences in rental affordability across cities.
3. Median Home Price
 - Captures baseline differences in housing affordability across cities.
4. Vacancy Rate
 - Measures housing pressure across cities.
5. Share of Remote-Work Occupations
 - Measures the prevalence of remote-work across cities.
6. Distance/Commute Time to San Francisco
 - Captures the attractiveness and accessibility of each city compared to San Francisco (migration patterns are likely shaped by commute feasibility).
7. Local Amenities
 - Includes education, transportation, shopping, safety, services, healthcare.
 - Controls for non-price factors that affect attractiveness.

Intervention (IV)

DLIT Implementation (binary: 1 = treatment, 0 = control)

This variable records whether a city receives the intervention.

- Treatment cities apply a DLIT only to new residents with incomes above the city median.
- Control cities do not adopt the tax and serve as the counterfactual.

- The treatment begins January 1, 2026 and ends December 31, 2031, providing a clear before/after comparison.

Outcomes (Dependent Variable)

1. Net Migration from San Francisco to Each City
 - Measures whether the dynamic local income tax reduces outflows from San Francisco into surrounding cities.
2. Median Rental Price
 - Measures yearly changes in rental affordability.
 - Supports hypothesis testing about whether treatment cities experience slower rent growth.
3. Median Home Price
 - Measures changes in property values.
 - Supports hypothesis testing about whether treatment cities experience slower property price appreciation.

8. Statistical Method

To determine the causal effect of introducing a DLIT on migration flows and housing affordability in the cities surrounding San Francisco, we will use a Difference in Differences (DiD) framework. This approach allows us to compare changes in the outcomes over time between the treatment cities (those implementing the DLIT) and the control cities, which isolates the effect of the intervention.

The key assumption behind DiD is parallel trends – that in the absence of the intervention, the treatment and control cities would have followed a similar rate in migration and housing affordability outcomes. We will use extensive, publicly available pre-intervention data about rent and housing prices, income distributions, and migration flows to validate this assumption by comparing outcome trends between the two groups.

We will run DiD models for the following primary outcomes:

1. Median Rent Price
2. Median Home Price
3. Net migration flows

$$Y_{it} = \alpha + (\beta \times \text{Treatment}_i) + (\gamma \times \text{Post}_t) + \delta \times (\text{Treatment}_i \times \text{Post}_t) + X_{it}\theta$$

Y_{it} : outcome variable for city i at time t

- Median rent price
- Median home price
- Net migration flows

Treatment_i : 1 if the city implemented the DLIT, 0 if not

Post_t : 1 for the years after 2026, 0 for before

δ : the DiD estimator; captures the causal effect of the tax

X_{it} : control covariates

- Median household income

- Vacancy rates
- Share of remote-work occupations
- Distance & commute time to San Francisco
- Local amenities

9. Potential Risks

- The legality of imposing a local income tax on new residents who are migrating from outside the established limits of a city, and not imposing local income tax on current residents.
- The study duration may be insufficient to observe the full effects of the treatment on the surrounding cities.
- The dynamic tax model, while bracketed to reference the current rate discrepancy at the onset of the study, may have larger variations at the end of the five year study.

10. Assumptions

- Assuming remote-workers are the high-income households moving away from the CBD to surrounding cities.
- We will assume that a remote-worker who moves during the study, will stay a remote worker throughout the study.
- Assuming California would allow the local income tax bracket systems for surrounding cities to only include new residents.
- Assuming all outflows from San Francisco are going to the surrounding cities.
- Median household income is not normally distributed. We will assume a lognormal distribution of income to assign a median.
- People moving out of San Francisco for financial arbitrage.

10. Deliverables

This study seeks to investigate the effects of imposing a DLIT on migration patterns between the CBD of San Francisco and the surrounding cities. Further investigation will focus on the resulting impact on housing affordability associated with the migration patterns outlined in the “Donut Effect”. The purpose of this study is to inform the Bay Area Housing Finance Authority on the effectiveness of using a DLIT on helping preserve and produce affordable housing across the San Francisco Bay Area. The imposition of this treatment is meant to counteract the effects of the “Donut Effect” by targeting an underutilized local income tax control mechanism to de-incentivize migration from high-income, high mobility, households seeking price arbitrage in housing costs. This study seeks to understand if a DLIT intervention would significantly decrease the population outflows from San Francisco into the surrounding cities, and stabilize the rising rent and home prices. Out of the 100 surrounding cities neighboring San Francisco, the 12 closest ‘donut’ cities were selected for the experiment. Simple random assignment was used to apply the treatment to the 12 cities, where each city

was randomly assigned a number, the list of cities was then sorted in ascending order, then half the group was assigned to the treatment, and the other half assigned to the control group.

The final deliverable will include a formal report to the Bay Area Housing Finance Authority. The report will contain a data analysis of the before and after state of migration and household costs on the control and treatment group using the Difference-in-Differences framework. In addition, the analysis will compare the net migration of the surrounding cities to the net migration of the San Francisco CBD to address our secondary stakeholder, the San Francisco Mayor's Office of Housing and Community Development. The report will detail the results of the study in reference to the null and alternative hypothesis, and determine if the dynamic local income tax implementation had a significant impact on migration patterns and cost of living. Details of the DLIT calculations will be provided, and how the dynamic tax model is applied to the sampling frame of the study.

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12. Statements of Contribution

Jordan Agresti: Worked on anticipated impact, problem statement, study design, assumptions, statistical methods

Santi Merida: Worked on main question, sub-questions, study design, potential risks

Justin Engwall: Worked on literature review, references, deliverables, assumptions, potential risks

Martin Dela Cruz: Worked on study design, sample size, hypotheses statistical method